제 8 장 인버터

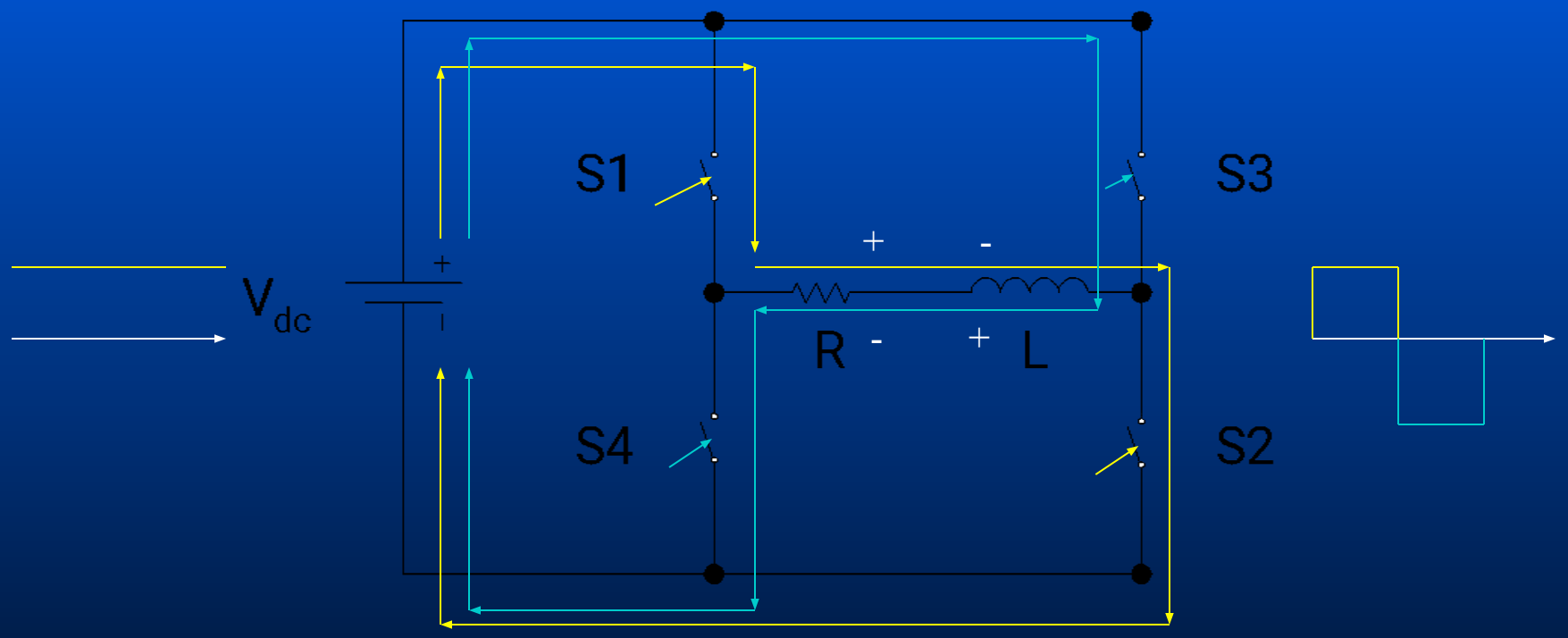
Inverter

DC→ AC

8.1 개요

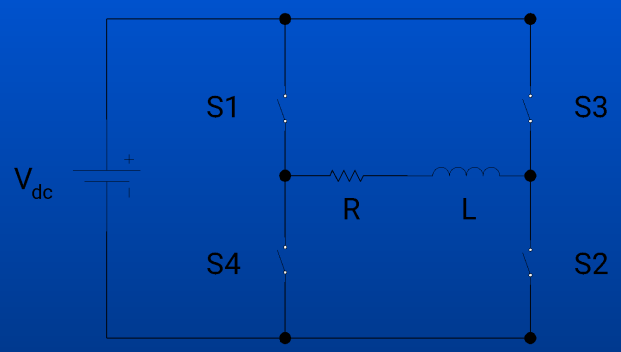
- DC를 이용하여 AC로 변환
- 용도
  - AC 모터속도제어
  - UPS(Uninterruptible Power Supply)
  - 태양광 발전
- 종류
  - 전브리지 변환기
  - 구형파 변환기
  - PWM 변환기

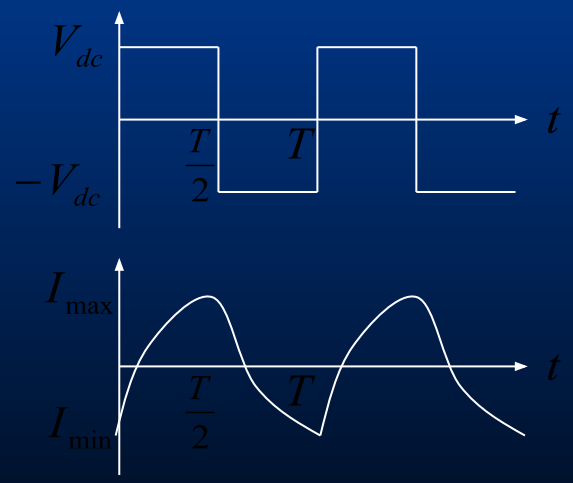
8.2 전브리지(Full Bridge)변환기



- DC 입력을 S1+S2와 S3+S4를 주기적으로 동작하여 AC로 변환

8.3 구형파(Square Wave)변환기





- S1, S2 ON일 때

$$L\frac{di_0}{dt}+Ri_0=V_{dc}$$ Laplace Transform

$$L\{SI(S)-i_0(+)\}+RI(S)=\frac{V_{dc}(S)}{S} \quad i_0(+)=I_{\min}$$

$$LSI(S)+RI(S)=\frac{V_{dc}(S)+SLI_{\min}}{S} \quad \div L$$

$$\left\{S+\frac{R}{L}\right\}I(S)=\frac{1}{L}\left\{\frac{V_{dc}(S)+SLI_{\min}}{S}\right\}$$

$$\therefore I(S)=\frac{\frac{1}{L}\{V_{dc}(S)+SLI_{\min}\}}{S\left\{S+\frac{R}{L}\right\}}$$

8.3 구형파(Square Wave)변환기(2)

$$I(S)=\frac{\frac{1}{L}\{V_{dc}(S)+SLI_{\min}\}}{S\left\{S+\frac{R}{L}\right\}}=\frac{A}{S}+\frac{B}{\left\{S+\frac{R}{L}\right\}}=\frac{A\left\{S+\frac{R}{L}\right\}+BS}{S\left\{S+\frac{R}{L}\right\}}$$

$$A \rightarrow S=0 \qquad \frac{R}{L}A=\frac{1}{L}V_{dc}(S), \qquad \therefore A=\frac{V_{dc}(S)}{R}$$

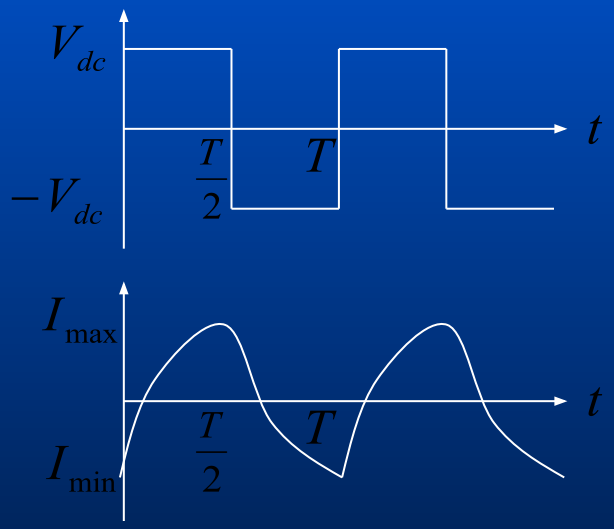
$$B \rightarrow S=-\frac{R}{L} \qquad -\frac{R}{L}B=\frac{1}{L}\left\{V_{dc}(S)-\frac{R}{L}LI_{\min}\right\}, \quad \therefore B=-\frac{1}{R}V_{dc}(S)+I_{\min}$$

$$I(S)=\frac{\frac{V_{dc}(S)}{R}}{S}+\frac{-\frac{1}{R}V_{dc}(S)+I_{\min}}{\left\{S+\frac{R}{L}\right\}}$$

Inverse Laplace Transform

$$i_0(t)=\begin{cases}\frac{V_{dc}}{R}+\left\{I_{\min}-\frac{V_{dc}}{R}\right\}e^{-\frac{R}{L}t} & at\ 0\le t\le\frac{T}{2},\ \because\ L^{-1}\left\{\frac{1}{(S-a)}\right\}=e^{-at}\\ -\frac{V_{dc}}{R}+\left\{I_{\max}+\frac{V_{dc}}{R}\right\}e^{-\frac{R}{L}\left(t-\frac{T}{2}\right)} & at\ \frac{T}{2}<t\le T\end{cases}$$

8.3 구형파(Square Wave)변환기(3)



$$i_0(t) = \frac{V_{dc}}{R} + \left\{ I_{min} - \frac{V_{dc}}{R} \right\} e^{-\frac{R}{L}t}$$

$$at \;\; t = \frac{T}{2}, \quad i_0(t) = I_{max}, \; (I_{min} = -I_{max})$$

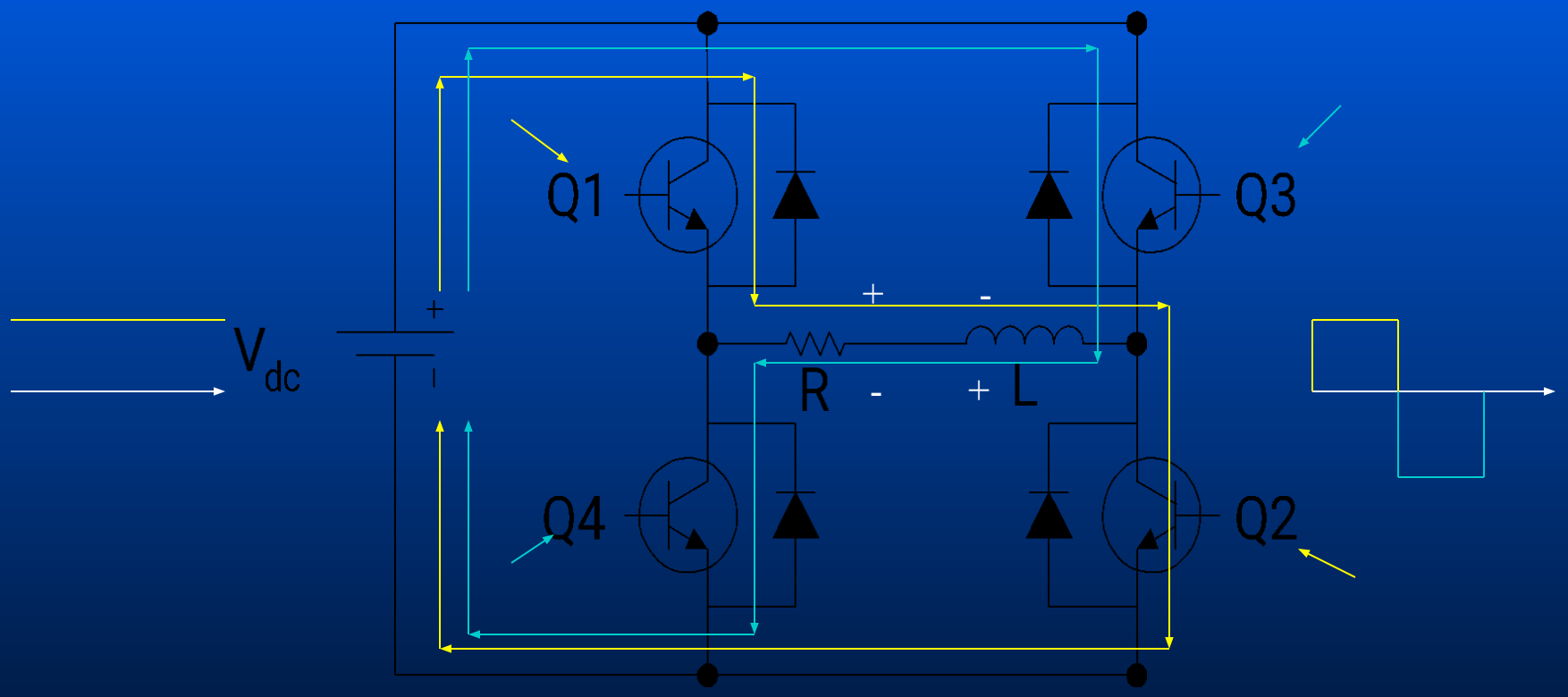
$$i_0\left(\frac{T}{2}\right) = I_{max} = \frac{V_{dc}}{R} + \left\{ -I_{max} - \frac{V_{dc}}{R} \right\} e^{-\left(\frac{T}{2\tau}\right)},$$

$$I_{max} + I_{max} e^{-\left(\frac{T}{2\tau}\right)} = \frac{V_{dc}}{R} - \frac{V_{dc}}{R} e^{-\left(\frac{T}{2\tau}\right)}$$

$$I_{max}\left\{ 1 + e^{-\left(\frac{T}{2\tau}\right)} \right\} = \frac{V_{dc}}{R}\left\{ 1 - e^{-\left(\frac{T}{2\tau}\right)} \right\}$$

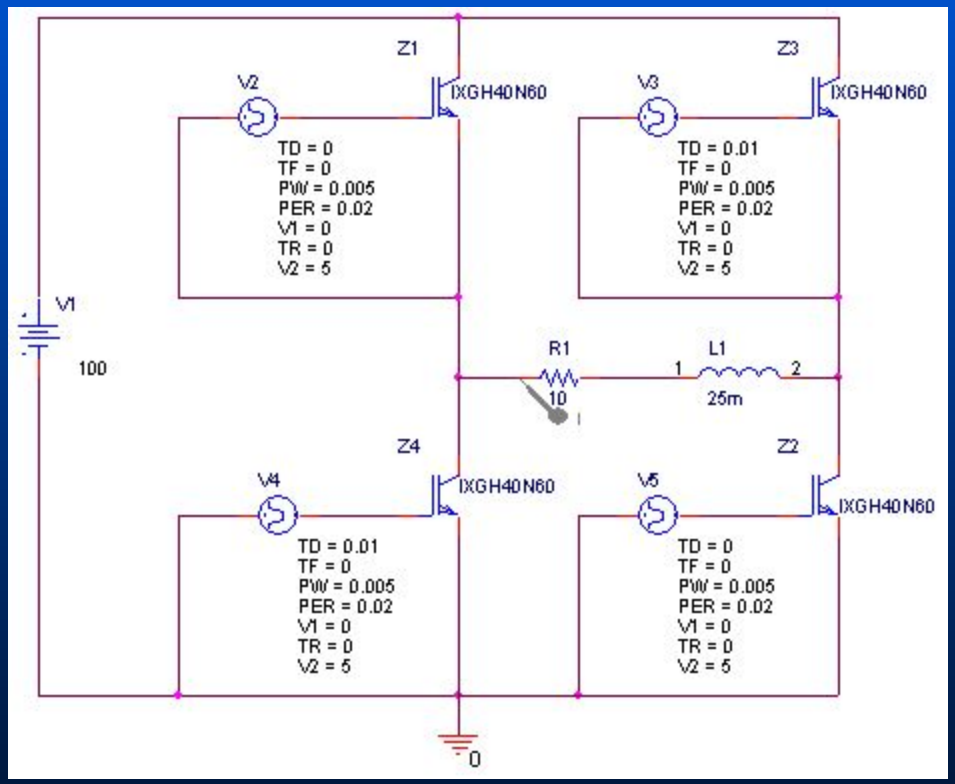
$$\therefore I_{max} = \frac{V_{dc}}{R} \frac{\left\{ 1 - e^{-\left(\frac{T}{2\tau}\right)} \right\}}{\left\{ 1 + e^{-\left(\frac{T}{2\tau}\right)} \right\}}$$

8.3 구형파 변환기: BJT 회로



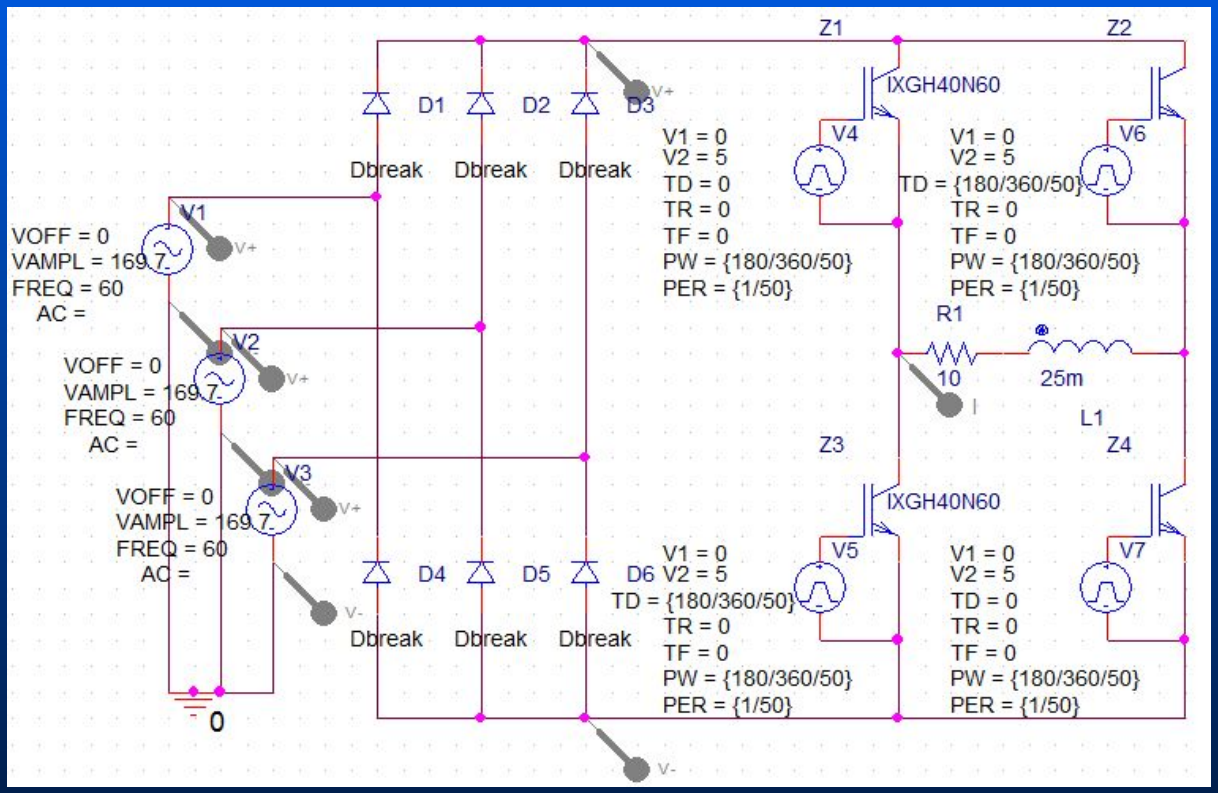
- S1+S2와 S3+S4를 주기적으로 동작 AC로 변환

예제 8-1 인버터



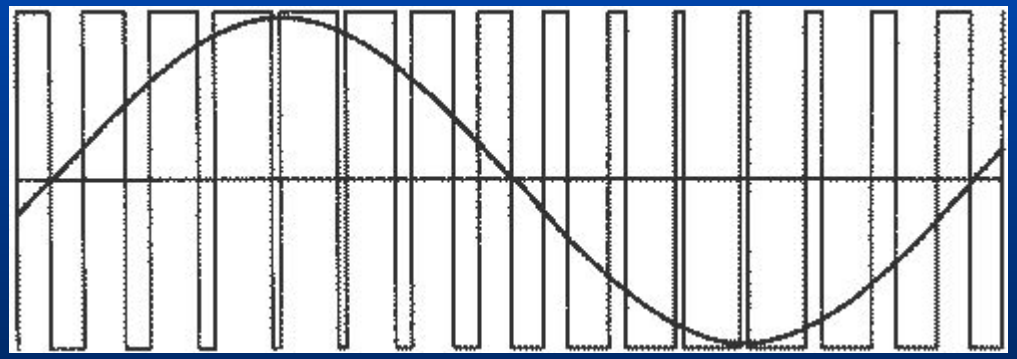
- *Parameter*
  - ✔ V_S=100 V
  - ✔ R=10 ohm
  - ✔ L=25 mH
  - ✔ f=50Hz (T=0.02 s)
  - ✔ PW<1/100(0.01 s)
  - ✔ Step
    =0 50 ms
- *Find*
  - ✔ 출력전압
  - ✔ 전류 최대, 최소값

예제 8-1 정류기가 있는 인버터



- *Parameter*
  - ✔ *V_S=170 V C=10 Mf R=10 ohm L=25 mH*
  - ✔ *f=50Hz (T=0.02 s) PW<1/100(0.01 s) Step : 0 50 ms b*
- *Find : 출력전압 전류 최대, 최소값*

8.10 PWM 변조 기술



- PWM(Pulse Width Modulation): 펄스폭 변조
- 펄스폭을 세분하여 동작시킴으로 정교한 정현파(sin wave)을 얻을 수 있음

8.10 PWM 기술: 변조비

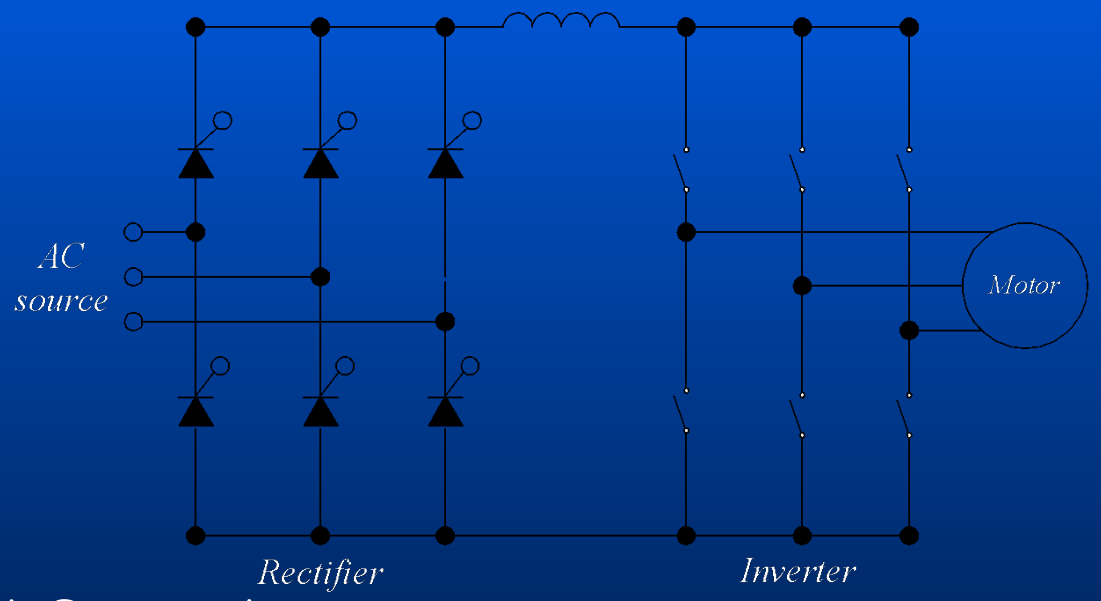
- 주파수 변조비(Frequency Modulation Ratio)

$$m_f = \frac{f_{carrier}}{f_{reference}}$$

- 크기 변조비(Amplitude Modulation Ratio)

$$m_a = \frac{V_{m \cdot reference}}{V_{m \cdot carrier}} \qquad if \quad m_a \le 1 \quad V_1 = m_a V_{dc}$$

8.17 Induction Motor 속도제어

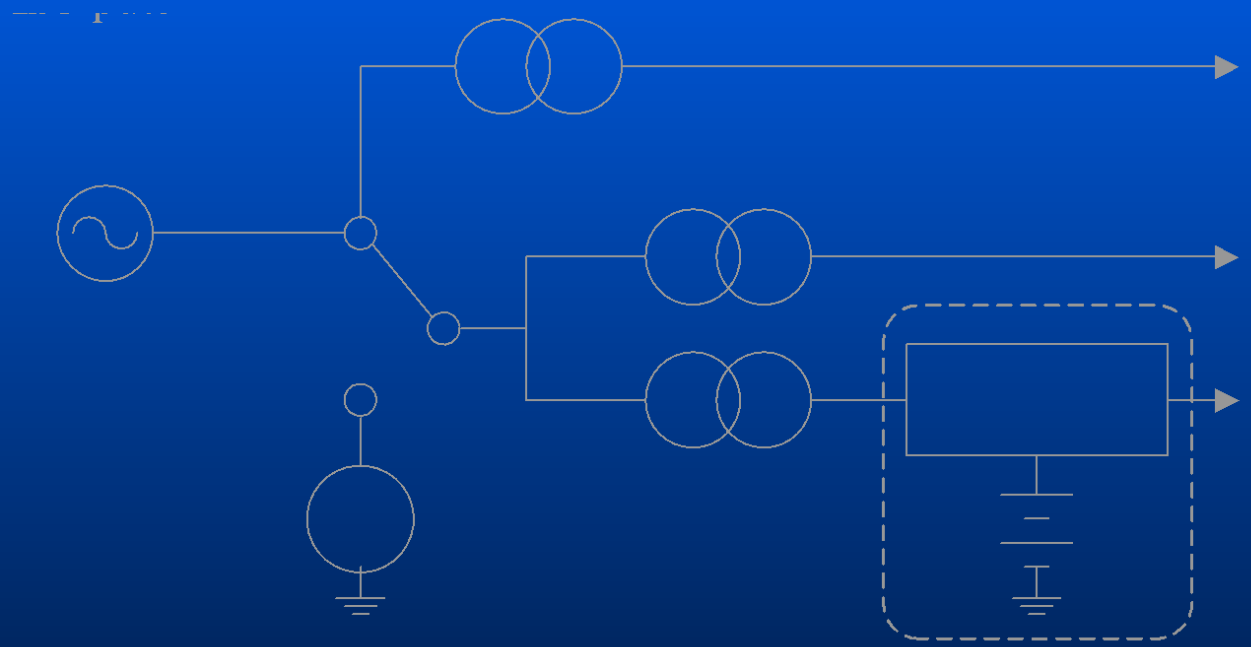


- Speed Control

$$N = \frac{120f}{P}(1-s)$$

- 회전수는 주파수에 비례
- 주파수 제어로 정밀한 회전수(속도) 제어가 가능
- 전동열차, elevator, 전기차등에 사용

UPS(무정전 전원공급장치)



- 정전시에 축전지에 저장된 DC를 INVERTER를 사용 AC로 변환하여 정전사고 방지
- UPS(Uninterruptible Power Supply)
- ESS(Energy Storage System): 에너지 저장장치

전력전자 종합

- AC → DC : 정류기(Rectifier)
- DC → AC : 인버터 (Inverter)
- AC → AC : AC-AC 컨버터 (AC-AC Converter)
- DC → DC : 초퍼 (DC chopper)
  - 승압 초퍼(Boost chopper)
  - 강압 초퍼(Buck chopper)
  - 강압 및 승압 초퍼(Buck-Boost chopper)

참고자료: Induction Motor 속도제어기

참고자료: UPS & ESS